Krishna Mula

+1 (408) 569-6676 | krishna.mula@sjsu.edu | krishmula.github.io | linkedin.com/krishna-mula | San Jose, CA, 95110

Education

San Jose State University San Jose, CA
Master of Science in Computer Science *Aug 2025 - May 2027*
Coursework: Distributed Systems, Database Systems, Cloud Computing, Machine Learning, NoSQL Database Systems.

Experience

PwC
Software Engineer *Aug 2022 - Jul 2025*

- Architected **RAG pipeline** for IT security compliance, embedding documents into ChromaDB for semantic retrieval and integrating **LLM inference** for automated validation against regulatory standards, **reducing review time by 70%**.
- Designed a **RabbitMQ-backed async ingestion architecture** that decoupled upload APIs from parsing, chunking, embedding, and vector indexing, **reducing latency by 30%** under bursty workloads.
- Engineered a centralized IT security knowledge base using LLM-driven extraction and retrieval workflows, automating **80% of manual source aggregation across 6 teams**.
- Built ETL pipeline using PyRFC to extract SM20/SM21 security audit logs from **5 production SAP systems**, automating a process that previously required **2+ hours of manual export and formatting** per report.
- Automated security controls execution on client SAP systems by codifying **25+ controls** as Flask + MongoDB REST APIs, ingesting data directly from SAP HANA and validating against security specifications, **cutting audit effort by 60%**.
- Deployed serverless REST APIs on Azure Functions and AWS Lambda with pre-warm triggers, **serving compliance reports across 15+ systems**, optimizing for 20-minute twice-daily burst windows.

PwC
Software Engineering Intern *Feb 2022 - Aug 2022*

- Built Nivo.js dashboards tracking security control pass/fail rates, compliance coverage, and historical trends, enabling teams to monitor compliance posture and prioritize which controls to automate next.
- Automated deployments with **Azure CI/CD** pipelines, reducing release cycles from **2 days to 4.5 hours**.

Projects

Aether | *Python · TCP Sockets · FastAPI · Docker SDK · OpenTelemetry · FastAPI · D3.js*

- Built a **distributed pub-sub messaging system** over raw TCP with a gossip-based multi-broker mesh, UUID deduplication, configurable fanout/TTL, and heartbeat-driven peer eviction.
- Implemented **Chandy-Lamport global snapshots** with replication to k peers; snapshots drive hybrid broker failover — fresh snapshot triggers full state recovery, stale snapshot triggers subscriber redistribution to least-loaded brokers.
- Built a **FastAPI orchestration control plane** over the Docker SDK with a pull-based subscriber assignment registry as the source of truth during recovery — same model as Kafka's consumer group coordinator.
- Instrumented with **OpenTelemetry structured logs**, **Prometheus metrics**, and provisioned **Grafana dashboards** for system overview and failover forensics; zero manual setup on make demo.
- Subscribers detect **broker failure via Ping/Pong**, reconnect with exponential backoff and jitter, and query the assignment API to self-heal without coordinated restarts.

Sero | *Python · FastAPI · AWS CDK · AWS Aurora Serverless · SQS · AWS Lambda*

- Built a **collaborative bill-splitting app** where users snap a receipt and roommates claim line items on their phones; deployed as a cross-platform PWA on iOS, Android, and web.
- Built an **async OCR pipeline** over SQS — decoupled upload from parsing via job queue, with a dead-letter queue for failed jobs and automatic retries for transient errors like timeouts.
- Provisioned the full stack with **AWS CDK** — API Lambda, SQS-triggered worker Lambda, one-shot migration Lambda — with Aurora Serverless behind **RDS Proxy** to pool connections so Lambda's bursty invocation pattern doesn't exhaust the database.

Technical Skills

Languages: Python, TypeScript, SQL, C++, Shell Scripting.
Frameworks: Flask, Node.js, FastAPI, React, Next.js.
Databases: PostgreSQL, MongoDB, Cassandra, Redis, SQLite, DynamoDB.
Observability: Prometheus, Grafana, Loki
Tools & Cloud: Git, Docker, AWS (EC2, S3, Lambda, CloudFront, SQS, DynamoDB, RDS, Amplify, CloudWatch), Azure Cloud (Azure Functions, Azure VM, Azure EntraID), GitHub Actions, Nginx, Firebase.